DroneVPS - Project Status

Marker-free GNSS-denied navigation on a 4.0 kg hexacopter. Target: 100-150 m AGL, 3-5 km out-and-back, no GNSS, no ground marker. Twenty-six flights across 19-25 October. Flight 245 is the first that worked: 634 seconds on the VPS, LOITER throughout, zero failsafe, 0.44 m at 38 m. The system is now limited by the map rather than by drift.

Overall

78%

86% was claimed on 23 October and withdrawn: the scoring behind it was circular. This is scored only against the raw receiver.

The phantom was a model error, and it is gone.

Derotation subtracted ONE centre-valued shift from a 77.7 degree frame. A rotation moves a point at radius r by f*theta*(1 + r^2/f^2) - +25% at the top edge, +65% at the side. Predicted residual 25.6%; measured 16-32% on two flights. Now removed from every track.

A tilted camera is not looking at the aircraft.

The image centre sits alt*tan(tilt) from nadir and the matcher reported it as the position. 19.4 m at 110 m and 10 degrees. All four sign pairs tried against the receiver; map error 11.18 -> 8.69 m.

Both matchers run, because neither covers the day.

XFeat took 78% of frames at 11:38 and 0 of 40 at 15:20 - the shadows had swung 100 degrees from the map's. NCC correlates gradient magnitude and does not care.

The map halved, and is now the limit.

Fix error 11.74 -> 5.64 m over five flights, and 2.89 m between 60 and 80. The closing rate is 0.192 m/s against a far slower drift, so position settles there.

An item is 100% only when it has been verified in flight.

1. Platform and plumbing

Item		Status / evidence
Airframe, compute, sensors	**95%**	ZD850 hexa, CUAV X7 Pro, Jetson Orin, OAK-D Pro. Camera mount lowered 23 Oct; FC sits 8 cm above the CG, camera 22 cm below.
Camera and frame rate	**98%**	Both monos at 25 fps, crop removed, 95.1% pairing. Loop holds 25.0 Hz; iterations over 60 ms were 47.7% at 110 m before the matcher got cheaper, 9.7% after.
Vibration	**96%**	p99 4.4 m/s2, no clipping, unchanged across every flight. Ruled out as a cause of anything.
MAVLink routing	**94%**	Unchanged. Probing the FC live is still forbidden - it changes the shared stream state and once killed the node's rangefinder.
Services and auto-start	**100%**	Unchanged.
Flight data recording	**97%**	79-column CSV plus frames at 4 Hz. Per-frame displacement, fix age, matcher disagreement and the gyro windows are all logged now.
Preflight diagnostics	**92%**	Unchanged. The lidar zero check still runs before arm.

2. Position estimation

Item		Status / evidence
Optical flow	**92%**	Per-track derotation since 25 Oct. flow_disagree_px p50 0.050 px, the two cameras agreeing to a twentieth of a pixel. The veto fires on 2.2% of frames.
Dead reckoning over distance	**85%**	0.44 m at 38 m over 90 s; 1.42 m at 82 m for the first two minutes. Unchanged method.
Gyro derotation	**90%**	The model is right now - the gain never was the problem. The coefficient sweep moved the phantom 0.026 to 0.024 and the integration window was bit-identical on 6400 frames.
Camera / IMU configuration	**95%**	Unchanged. 169-176 Hz delivered, gravity removed by the BNO fusion, verified mean near zero.
Mount alignment	**90%**	Camera yaw offset -1.95 deg, measured on two flights (-2.03 and -1.87, spread 0.85) and the sign settled by replay.
Map matching	**80%**	159 fixes on flight 245 against a previous best of 70. Error p50 5.64 m; 2.89 m at 60-80 m, 11.37 m above 95. Both matchers run and their agreement predicts accuracy at corr +0.928.
Sensor fusion	**72%**	fix_sigma_m is still 20 while the map now measures 5.64 m, so the Kalman gain is 0.068 and a 5.6 m residual pulls 0.38 m. The correction written for this was reverted.
Velocity to the FC	**55%**	The phantom is far smaller but the deadband is a hard edge and the 40-frame baseline carries 1.6 s of lag. Those are faults 1 and 2 on the next page.
Heading / yaw	**55%**	Unchanged. The compass is still the only yaw source and the map corrects its bias.
Altitude	**45%**	The barometer reads 5-7 m high - 1.07x at 100 m, 1.72x at 10. Velocity is pixels x altitude / focal, so that ratio is the velocity error. The lidar is logged and flown on by nothing.
Position injection as GPS2	**90%**	Unchanged. GPS_INPUT as GPS2, the FC switches the primary, and nothing of ours reaches it while the switch is on the receiver.

3. Flight controller integration

Item		Status / evidence
EKF origin and home	**100%**	From the receiver at arm. The one flight that set it by hand from SCR_USER1/2 tracked worse and was discarded.
Manual RC GPS1/GPS2 switch	**100%**	Unchanged, and the cache that once swallowed the pilot's command is still gone.
Actually flying on the VPS	**85%**	634 s on flight 245 with no failsafe, against two to four ALT_HOLDs on every earlier flight. Held at 38, 82 and 108 m.
Smoothness	**35%**	The sent speed crosses zero 307 times in 634 s and the commanded tilt saturates at 15 degrees. This is what the pilot feels.
Autonomous mission	**40%**	Not attempted on the VPS since the rebuild.
Mission speed ceiling	**70%**	LOIT_SPEED_MS 3, WP_SPEED 4, both reduced from the 4.7 defaults and confirmed still reduced.
Behaviour on RC loss	**100%**	Unchanged.

4. What changed since the 23 October report

Item		Status / evidence
Per-track derotation	**100%**	FLOWN. Rotation removed from every track before the fit. flow_disagree_px 2.32 -> 0.050 px, the veto 95% -> 2.2%, velocity stale 41% -> 0.2%.
Both cameras derotated	**100%**	FLOWN. The first version did the primary only, so the gap between the cameras became the rotation and the veto ate 95% of flight 163. One call missing one keyword.
Tilt correction on the fix	**100%**	FLOWN. The answer moved from the image centre to under the aircraft. 11.18 -> 8.69 m and 9.86 -> 6.83 m. Signs measured, not derived.
NCC restored, both matchers	**100%**	FLOWN. matcher = "both". NCC needs a window larger than the frame footprint or it starves - that cost 0 of 40 fixes until it was found.
Camera yaw offset -1.95	**100%**	FLOWN. The first attempt tested the wrong sign and read it as harmful.
Map fix uses its own prior	**100%**	FLOWN. The residual was measured against the current position, folding 0.3-0.95 m of travel into a 0.49 m correction. The pilot described this before it was found.
Fix rate limit 5 m/s	**100%**	FLOWN. Reported honestly: on the fixes already flown it caught neither of the two worse than 25 m.
step_dn / step_de / step_dt_s	**100%**	FLOWN. The per-frame displacement is written down, which is what made the baseline sweep possible at all.
fix_age_s, fix_disagree_m	**100%**	FLOWN. Round trip 0.19-0.41 s, and how far the two matchers landed apart.
Deadband as a ramp	**0%**	REVERTED, commit 5066572. Zero at the threshold, full value at 1.5x it. Written for the jerking.
flow_vel_frames 40 -> 30	**0%**	REVERTED, commit e626f42. Re-swept after the derotation fix: lower phantom AND less lag. Written for the toilet bowl.
fix_sigma_m 20 -> 8	**0%**	REVERTED, commit e626f42. The map measures 5.64 m, not the 20 m of 9 October.
Lidar correcting the baro	**0%**	REVERTED, commit e626f42. A learned offset, never a replacement, so a bad lidar could not stop the aircraft flying.

The last four were written, flown once, and reverted at the pilot's instruction. That flight tracked worse on the map - but the origin was also set by hand from SCR_USER1/2 on the same flight, so four things changed at once and nothing is separable. The FC log was deleted and the tree is back to exactly what flight 245 flew. Reintroduce them ONE AT A TIME.

5. What is still wrong

Item		Status / evidence
The aircraft jerks	**30%**	The sent speed crosses zero 307 times in 634 s and the deadband zeroes 39.9% of rows. A hard threshold: below it the FC is told exactly 0, above it the full value, so every crossing is a step. Commanded tilt saturates at 15 deg, rates to 56.9 deg/s. Map fixes ruled out - 16% coincidence against a 15% base rate. It only became the limiting fault because the phantom shrank onto the threshold.
Toilet bowl at 35 m	**30%**	The bearing from its own centre swept 754 degrees in 24 s while the yaw moved 9.6 - circling, not turning. Radius p50 3.72 m. Period 11.4 s; 2*pi times the 40-frame baseline's 1.6 s lag is 10.1 s. The first hold of the same flight swept 18 degrees, because there the deadband was zeroing 68% of the output and the loop was open.
The barometer reads high	**40%**	Against the lidar: 1.07x at 100 m, 1.17x at 40, 1.29x at 20, 1.72x at 10. The same climbing and descending, so an offset and not hysteresis. At touchdown the lidar read 0.60 m and the altitude in use read 4.2. That ratio is the velocity error: 72% at 10 m, which is where the aircraft was circling.
The map above 95 m	**45%**	11.37 m against 2.89 m between 60 and 80. Fly the map work low until this is understood.
The error creeps	**60%**	1.4 m to about 10 m at 82 m over a few minutes, then it holds. Consistent with the map's own accuracy plus correction lag, so it may be what map-limited looks like - but it has not been separated.

6. Closed by measurement - do not reopen

Item	Status / evidence
Scale / focal error	0.79-1.02 against truth with both sides smoothed equally. The "2.22x" that started the hunt was an artefact.
The camera accelerometer	Never free-runs. vel_age_s p99 0.18 s against a 0.6 s timeout, zero rows over it.
The gyro integration window	Bit-identical on 6400 frames. The drain window already was the frame window.
Tracker noise	98% of the per-frame step is common to both cameras, so it is real image motion. Each camera's own noise is 0.035 px against a 1.9 px step.
The derotation coefficient	Swept 14.80 down to 13.40: the phantom moved 0.026 to 0.024.
Aircraft oscillation	The step's energy sits at 0.02-0.17 Hz; the attitude oscillates at 0.26, 0.86, 1.03.
The camera-to-FC lever arm	0.157 m/s at 30 deg/s against a 1.4-2.5 m/s phantom. Real, 8%, not the cause.
Raising max_correction_m	Flown 3 Oct: the hold fell from 11-14 s to 4.5-9.0 s. Inert anyway - the gain binds first.
Height-scaled velocity baseline	Flown log 229: 11.1 m in 24 s against 0.9 m on a flat baseline.
Declaring a larger sigma	Flown and reverted: 12x further at 31 m. This aircraft has no other horizontal source.
Map window radius	30 and 40 m collapse the fix rate, 50 and 66 are identical, 90 is within noise.
Edge / corner filtering	Makes it WORSE, 12.11 -> 12.91 m. A 12 m tree leans 9.7 m at the image edge and the ortho was built from other angles, so flat ground is the honest target.
Nadir cropping	No consistent gain and the fix rate collapses.

Four of this report's earlier conclusions were withdrawn, and all four came from the same mistake. Score position only against GPS_RAW_INT, never against the EKF's fused output, which on VPS is our own injection returning. Filter those columns first - they contain rows 23 km out. Smooth both sides of any comparison equally. And before blaming a code change for a regression, replay that flight's own frames with the change on and off.